

Computational Social Contract (CSC): Protocol-Level Collective Alignment in Heterogeneous Multi-Agent Systems via an Algorithmic Veil of Ignorance

Anonymous Authors
Affiliation
Address
email

Abstract

As AI systems transition from single agents to interacting collectives, alignment must extend from individual intent-following to *collective* objectives that remain stable under heterogeneity, scarcity, and adversarial deviations. We introduce the **Computational Social Contract (CSC)**, a protocol that (i) learns a *contract policy* under an identity-blind **Algorithmic Veil of Ignorance (AVoI)** and a welfare objective that discourages near-zero outcomes, and (ii) stabilizes identity-aware execution via **KL grounding** and an **arbitration layer** trained against evasive deviations. We propose *Silicon Commune*, a scalable heterogeneous resource dilemma benchmark with controlled shock families and Byzantine stress tests. Across strong CTDE baselines (VDN/QMIX/COMA and PPO-style centralized-critic variants), CSC improves equity (Gini ↓, Jain ↑) and robustness (shock survival ↑, Byzantine resilience ↑) while maintaining competitive throughput.

1 Introduction

Multi-agent systems are increasingly deployed as *collectives*: swarms of agents sharing resources, negotiating constraints, and affecting each other’s outcomes [4, 25]. In such settings, naive efficiency maximization can produce *polarization* and *fragility*: strong agents exploit weak ones, drive some returns toward zero, and create single points of failure that collapse under shocks or strategic deviations [8, 9, 16, 23]. This motivates *collective alignment*: a design goal where cooperative behavior remains equitable and robust under heterogeneity and distribution shift [3, 28].

A natural approach is to encode social welfare in the objective [31, 35]. However, welfare-only training is insufficient when execution policies can deviate after learning, or when learning exploits identity-index shortcuts that do not generalize. Moreover, in heterogeneous populations, identity features (endowments, capabilities) create incentives to specialize policies to privileged roles. This gap suggests a protocol view: separate *contract formation* (how the group agrees on a rule) from *contract execution* (how individual policies behave when identities are real) [29, 33].

Our approach. We propose **Computational Social Contract (CSC)**, a protocol with two phases: (i) **Contract Formation** learns a contract policy under an **Algorithmic Veil of Ignorance (AVoI)** that suppresses identity-index information during learning, inspired by the “original position” [27]; (ii) **Execution Grounding** trains identity-aware execution policies regularized toward the contract and stabilized by an **arbitration layer** that penalizes detected violations and is trained against evasive deviations [10, 19].

Contributions.

- **Protocol-level formulation.** We formulate collective alignment as contract formation under AVoI plus identity-aware execution grounded by KL regularization and arbitration.
- **Implementable AVoI operators.** We define a family of masking operators (permutation/partial/noise) that induce symmetry and reduce identity-index shortcutting.
- **Benchmark and evaluation axes.** We introduce *Silicon Commune* with controlled shock processes and Byzantine stress tests, and evaluate efficiency–equity–robustness jointly.

2 Related Work

Cooperative MARL and CTDE. Our baselines follow the CTDE paradigm [2, 22], including value decomposition (VDN [34], QMIX [26]) and counterfactual policy gradients (COMA [6]). We build on Markov games and stochastic games foundations [17, 32].

Social dilemmas and fairness in MARL. Sequential social dilemmas formalize tensions between individual incentives and group welfare [16]. In inequity settings, intrinsic or shaped objectives can improve cooperation [9, 12]. CSC differs by separating *contract learning* from *execution enforcement*, and by explicitly suppressing identity-index information during contract formation.

Welfare objectives and resource allocation fairness. Nash bargaining and Nash social welfare connect efficiency with fairness by penalizing near-zero outcomes [13, 20]. Classic inequality measures include the Gini coefficient [7] and Jain’s index [11], and proportional fairness arises in network utility maximization [14, 18]. CSC uses welfare in contract formation and complements it with enforcement against deviations.

Robustness and adversarial deviations. Robust RL and adversarial training study stability under disturbances and adversaries [1, 24]. In multi-agent systems, malicious or Byzantine behaviors are a classical threat model [15]. CSC operationalizes robustness by training an arbitration detector against evasive deviation policies and evaluating across shock families and Byzantine stress tests.

3 Problem Formulation

We consider an n -agent Markov game with global state s_t , local observations $o_{i,t}$, joint action $a_t = (a_{1,t}, \dots, a_{n,t})$, and per-agent reward $r_{i,t}$ [17, 32]. Each agent has an endowment/identity vector $\phi_i \in \mathbb{R}^d$ (capabilities, capacities, or costs) that affects rewards and transitions.

Objective: Nash Social Welfare. We adopt an ϵ -stabilized Nash Social Welfare (NSW) objective:

$$\mathcal{W}_{\text{NSW}}(\pi) = \frac{1}{n} \sum_{i=1}^n \log(\epsilon + \mathbb{E}_{\pi}[G_i]), \quad (1)$$

where $G_i = \sum_{t=1}^T \gamma^{t-1} r_{i,t}$ is the episodic return. NSW is a standard welfare criterion that balances efficiency and discourages near-zero outcomes [13, 20].

Proposition 1 (NSW discourages near-zero outcomes). *Holding total return fixed, the NSW objective (1) penalizes allocations that drive any $\mathbb{E}_{\pi}[G_i]$ toward zero more strongly than linear welfare, inducing an implicit inequality aversion.*

4 Method: Computational Social Contract (CSC)

CSC separates **contract formation** from **execution**. Contract formation learns a symmetric contract policy under identity masking. Execution trains identity-aware policies regularized toward the contract and stabilized by an arbitration layer.

4.1 Algorithmic Veil of Ignorance (AVoI)

Definition 1 (AVoI operator family). Let $\Phi = \{\phi_1, \dots, \phi_n\}$ be the set of identity/endowment vectors. AVoI produces masked identities $\bar{\phi}_i$ using a masking kernel $\mathcal{T}_{\eta}$:

$$\bar{\phi}_i \sim \mathcal{T}_{\eta}(\phi_i; \Phi),$$

where η controls masking strength. The masked observation is $o'_{i,t} = (o_{i,t}^{env}, \bar{\phi}_i)$.

We implement three AVoI instances: (i) **Permutation AVoI**: sample $\sigma \sim \text{Unif}(\mathcal{S}_n)$ and set $\bar{\phi}_i = \phi_{\sigma(i)}$; (ii) **Partial AVoI**: dropout mask $m \sim \text{Bernoulli}(1 - \eta)^d$ and $\bar{\phi}_i = m \odot \phi_i$; (iii) **Noisy AVoI**: $\bar{\phi}_i = \phi_i + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$.

Proposition 2 (Permutation invariance of the masked objective). *Let $J(\pi) = \mathbb{E}_{\sigma}[\mathcal{W}_{\text{NSW}}(\pi; \sigma)]$ denote NSW under permutation AVoI. Then $J(\pi)$ is invariant to relabeling agents, encouraging symmetric contract policies.*

4.2 Two-phase training

Phase I: Contract formation. Agents learn π_{contract} on masked observations to maximize NSW:

$$\max_{\theta_c} \mathcal{W}_{\text{NSW}}(\pi_{\text{contract}}(\cdot; \theta_c)) \quad \text{under AVoI.}$$

Algorithm 1 CSC (AVoI Contract + Grounding + Arbitration)

- 1: Initialize contract policy $\pi_{\text{contract}}(\theta_c)$, execution policy $\pi_{\text{exec}}(\theta_e)$, proposer q_ψ , detector d_ω .
 - 2: **for** each iteration **do**
 - 3: Sample identity set $\Phi = \{\phi_i\}$ and AVoI mask (e.g., permutation σ).
 - 4: **Phase I (Contract)**: rollout masked obs $o'_{i,t}$, update θ_c to maximize NSW (1).
 - 5: **Proposer/Detector**: update q_ψ to find high-gain evasive deviations; update d_ω via balanced BCE (Appendix D).
 - 6: **Phase II (Execution)**: rollout real obs $o_{i,t}$, shape reward (3), update θ_e with objective (2).
 - 7: **end for**
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Phase II: Execution grounding. Agents learn identity-aware execution policies π_{exec} on real identities, regularized toward the contract:

$$\max_{\theta_e} \sum_i \mathbb{E}[G_i] - \beta \sum_i \mathbb{E}[D_{\text{KL}}(\pi_{\text{exec}}(\cdot|o_{i,t}) \parallel \pi_{\text{contract}}(\cdot|o'_{i,t}))], \quad (2)$$

where β controls contract adherence.

4.3 Arbitration layer: deviation proposer and detector

CSC deters deviations with an arbitration detector $d_\omega(o, a) \in [0, 1]$ trained to discriminate contract-compliant actions from deviations. A deviation proposer $q_\psi(\cdot|o)$ searches for profitable/evasive deviations. During execution, CSC shapes reward:

$$r_{i,t}^{\text{shaped}} = r_{i,t} - \kappa \cdot \mathbb{I}[d_\omega(o_{i,t}, a_{i,t}) \geq \tau], \quad (3)$$

with penalty scale κ and threshold τ calibrated at a fixed false-positive rate (Appendix D). Reward shaping is grounded in classical theory [21] but here is tied to learned violation detection.

Lemma 1 (Deviation success bound (informal)). *If the arbitration penalty scale κ exceeds the maximal achievable deviation advantage over horizon H , then the successful deviation rate (SDR) is bounded by the detector’s false-negative rate (FNR). Formal definitions and measurement protocol are in Appendix A.*

5 Experiments

5.1 Benchmark: Silicon Commune (main summary)

We introduce *Silicon Commune*, a heterogeneous resource dilemma designed to stress-test collective alignment under scarcity. Each episode has shared resource R_t and heterogeneous endowments $\phi_i = (c_i, m_i)$. Agents choose extraction requests $a_{i,t} \in [0, 1]$; realized allocations are proportional to attempted extraction under pool availability. Rewards use diminishing returns $u_i(e) = \log(1 + c_i e)$, plus action cost. Resource shocks follow Bernoulli/Poisson/Markov families (Appendix B), and Byzantine stress tests randomly replace a subset of agents with malicious policies (Appendix C).

Table 1: Silicon Commune main setting (full grids in Appendix).

Parameter	Value
Agents n	100
Episode length T	200
Regen ρ	1.0
Strong ratio	0.1
Strong (c, m)	$[5, 10] \times [5, 10]$
Weak (c, m)	$[1, 2] \times [1, 2]$
Action bins K	11
Masking strength η	0.8

Table 2: **Main results on Silicon Commune.** Mean $\pm$ std over 5 seeds. Resilience AUC averages survival over shock sweep grids (Appendix B).

Method	Throughput $\uparrow$	Gini $\downarrow$	Jain $\uparrow$	Resilience AUC $\uparrow$
PPO-CTDE (centralized critic)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
VDN [34]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
QMIX [26]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
COMA [6]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
CSC (Ours)	[TODO:]	[TODO:]	[TODO:]	[TODO:]

5.2 Baselines and protocol

We compare against CTDE baselines: VDN [34], QMIX [26], COMA [6], and a PPO-style centralized-critic baseline (clipped policy gradient with GAE) [30]. We match interaction budgets, parameter counts, and trial budgets (Appendix F).

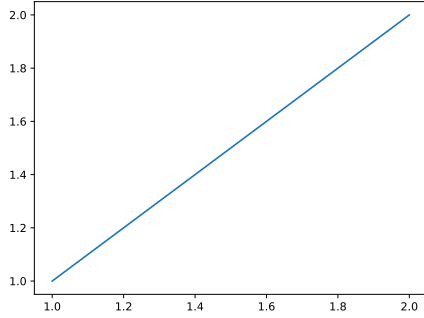
5.3 Metrics

We report: **Efficiency** (Throughput, NSW), **Equity** (Gini [7], Jain [11]), **Robustness** (Shock survival / Resilience AUC, Byzantine resilience), and **Deviation resistance** (DVR/SDR). Definitions and computation details appear in Appendix A.

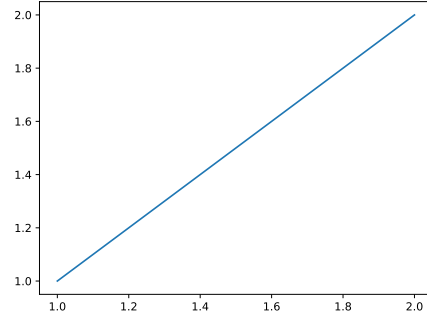
6 Results

6.1 Main results

CSC improves equity and robustness while maintaining competitive efficiency. Table 2 reports throughput, inequality metrics, and Resilience AUC over shock sweeps. Learning curves (NSW, Gini) appear in Figure 1.



(a) NSW over training steps (mean $\pm$ CI).



(b) Gini over training steps (mean $\pm$ CI).

Figure 1: Learning curves. CSC improves NSW while reducing inequality.

Table 3: **Ablations and deviation robustness.** SDR/DVR definitions: Appendix A.

Variant	Gini $\downarrow$	Res. AUC $\uparrow$	SDR $\downarrow$	DVR $\downarrow$
CSC (Full)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
w/o AVoI	[TODO:]	[TODO:]	[TODO:]	[TODO:]
w/o KL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
w/o Arbitration	[TODO:]	[TODO:]	[TODO:]	[TODO:]

6.2 Ablations and deviation robustness

Table 3 shows that removing AVoI, KL grounding, or arbitration degrades at least one axis among equity, robustness, and deviation resistance.

6.3 Shock and Byzantine robustness

Figure 2 visualizes resilience across shock parameters. Full sweep grids and Byzantine heatmaps appear in Appendix B and Appendix C.

7 Discussion and Limitations

Scope control. All claims are restricted to Markov-game collectives splitting finite resources and do not assume human-level semantics.

Limitations. Welfare choice: NSW discourages near-zero outcomes but can be sensitive when returns approach zero; we stabilize with ϵ and report Gini/Jain. **Masking realism:** AVoI assumes identity signals can be suppressed during contract learning; partial/noisy variants mitigate but do not eliminate this constraint. **Enforcement:** arbitration depends on detector calibration; we fix FPR and report ROC/AUC (Appendix D), but formal auditing

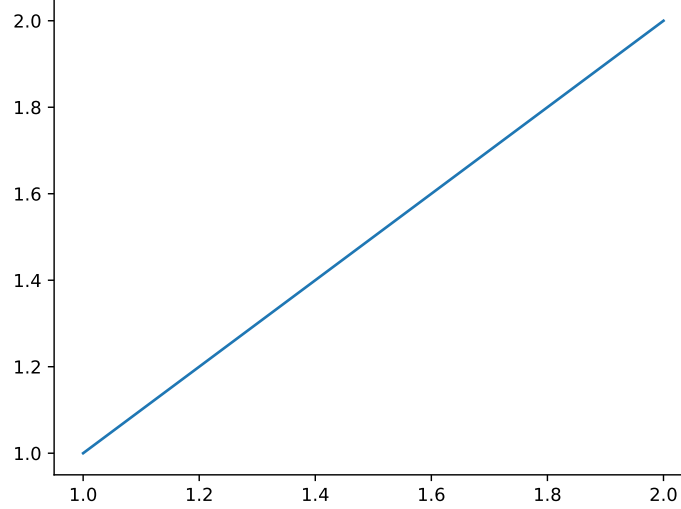


Figure 2: Resilience under resource shocks (survival probability). Full grids and AUC computation: Appendix B.

guarantees remain open. **Scaling:** arbitration introduces overhead; scaling and wall-clock are reported in Appendix.

8 Conclusion

We presented **Computational Social Contract (CSC)**, a protocol for collective alignment in heterogeneous multi-agent systems. CSC learns a symmetric contract under an Algorithmic Veil of Ignorance and stabilizes identity-aware execution via KL grounding and arbitration trained against evasive deviations. On *Silicon Commune*, CSC improves equity and robustness with competitive efficiency.

References

- [1] Vahid Behzadan and Arslan Munir. Vulnerabilities of deep reinforcement learning to policy induction attacks. In *International Conference on Machine Learning and Data Mining in Pattern Recognition (MLDM)*, 2017.
- [2] Daniel S. Bernstein, Robert Givan, Neil Immerman, and Shlomo Zilberstein. The complexity of decentralized control of markov decision processes. In *Proceedings of the 18th Conference on Uncertainty in Artificial Intelligence (UAI)*, 2002.
- [3] Brian Christian. *The Alignment Problem: Machine Learning and Human Values*. WW Norton & Company, 2020.

- [4] Allan Dafoe, Yoram Bachrach, Gillian Hadfield, Eric Horvitz, Kate Larson, and Thore Graepel. Cooperative AI: Machines must learn to find common ground. *Nature*, 593: 33–36, 2021.
- [5] Tom Fawcett. An introduction to roc analysis. *Pattern Recognition Letters*, 27(8): 861–874, 2006.
- [6] Jakob Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. Counterfactual multi-agent policy gradients. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2018.
- [7] Corrado Gini. Variabilità e mutabilità. *Studi Economico-Giuridici della R. Università di Cagliari*, 3:3–159, 1912.
- [8] Garrett Hardin. The tragedy of the commons. *Science*, 162(3859):1243–1248, 1968.
- [9] Edward Hughes, Joel Z. Leibo, Matthew Phillips, Karl Tuyls, Edgar Duéñez-Guzmán, et al. Inequity aversion improves cooperation in multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [10] Leonid Hurwicz. The design of mechanisms for resource allocation. *American Economic Review*, 63(2):1–30, 1973. Classic mechanism design.
- [11] Raj Jain, Dah-Ming Chiu, and William Hawe. A quantitative measure of fairness and discrimination for resource allocation in shared computer systems. *DEC Research Report TR-301*, 1984.
- [12] Natasha Jaques, Angeliki Lazaridou, Edward Hughes, Caglar Gulcehre, Pedro Ortega, et al. Social influence as intrinsic motivation for multi-agent deep reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2019.
- [13] Ehud Kalai and Meir Smorodinsky. Other solutions to the nash bargaining problem. *Econometrica*, 43(3):513–518, 1975.
- [14] Frank P. Kelly. Charging and rate control for elastic traffic. *European Transactions on Telecommunications*, 8(1):33–37, 1997.
- [15] Leslie Lamport, Robert Shostak, and Marshall Pease. The byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, 4(3):382–401, 1982.
- [16] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. Multi-agent reinforcement learning in sequential social dilemmas. In *Proceedings of the 16th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS)*, 2017.
- [17] Michael L. Littman. Markov games as a framework for multi-agent reinforcement learning. *Proceedings of the 11th International Conference on Machine Learning (ICML)*, 1994.
- [18] Jeonghoon Mo and Jean Walrand. Fair end-to-end window-based congestion control. *IEEE/ACM Transactions on Networking*, 8(5):556–567, 2000.
- [19] Roger B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1): 58–73, 1981.

- [20] John F. Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.
- [21] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. *Proceedings of the 16th International Conference on Machine Learning (ICML)*, 1999.
- [22] Frans A. Oliehoek and Christopher Amato. A concise introduction to decentralized POMDPs. In *SpringerBriefs in Intelligent Systems*. Springer, 2016.
- [23] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, UK, 1990.
- [24] Lerrel Pinto, James Davidson, Rahul Sukthankar, and Abhinav Gupta. Robust adversarial reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2017.
- [25] Iyad Rahwan, Manuel Cebrian, Nick Obradovich, Josh Bongard, Jean-François Bonnefon, Cynthia Breazeal, Jacob W. Crandall, Nicholas A. Christakis, Iain D. Couzin, Matthew O. Jackson, et al. Machine behaviour. *Nature*, 568(7753):477–486, 2019.
- [26] Tabish Rashid, Mikayel Samvelyan, Christian Schroeder, Gregory Farquhar, Jakob Foerster, and Shimon Whiteson. Qmix: Monotonic value function factorisation for deep multi-agent reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, 2018.
- [27] John Rawls. *A Theory of Justice*. Harvard University Press, Cambridge, MA, 1971.
- [28] Stuart Russell. *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking, 2019.
- [29] Tuomas Sandholm. Distributed rational decision making. *Multiagent Systems: A Modern Approach to Distributed Artificial Intelligence*, 1999.
- [30] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.
- [31] Amartya Sen. *On Economic Inequality*. Oxford University Press, 1973.
- [32] Lloyd S. Shapley. Stochastic games. *Proceedings of the National Academy of Sciences*, 39(10):1095–1100, 1953.
- [33] Yoav Shoham. *Agent-Oriented Programming*. Artificial Intelligence, 1993.
- [34] Peter Sunehag, Guy Lever, Audrunas Gruslys, Wojciech Czarnecki, Vinicius Zambaldi, et al. Value-decomposition networks for cooperative multi-agent learning. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2018.
- [35] H. Peyton Young. *Equity: In Theory and Practice*. Princeton University Press, 1994.

A Metric Definitions

A.1 Throughput and Returns

We report system throughput as total realized extraction per episode:

$$\text{Throughput} = \sum_{t=1}^T \tilde{E}_t.$$

Episodic return:

$$G_i = \sum_{t=1}^T \gamma^{t-1} r_{i,t}^{shaped}.$$

A.2 Nash Social Welfare (NSW)

We use an ϵ -stabilized form:

$$\mathcal{W}_{\text{NSW}}(\pi) = \frac{1}{n} \sum_{i=1}^n \log (\mathbb{E}_{\pi}[G_i] + \epsilon), \quad \epsilon = 10^{-3}.$$

A.3 Gini and Jain

Gini:

$$\text{Gini}(x) = \frac{\sum_{i=1}^n \sum_{j=1}^n |x_i - x_j|}{2n^2 \bar{x} + \epsilon_G}, \quad \bar{x} = \frac{1}{n} \sum_i x_i.$$

Jain:

$$\text{Jain}(x) = \frac{(\sum_{i=1}^n x_i)^2}{n \sum_{i=1}^n x_i^2 + \epsilon_J}.$$

A.4 Collapse and Resilience

Collapse:

$$\text{Collapse} \iff \exists t \leq \alpha T : R_t \leq \epsilon_R, \quad \alpha = 0.3, \epsilon_R = 10^{-3}.$$

Resilience:

$$\text{Resilience} = \Pr[\neg \text{Collapse}].$$

Resilience AUC averages survival over a parameter grid (Appendix B).

A.5 DVR and SDR

DVR:

$$\text{DVR} = \frac{1}{nT} \sum_{i=1}^n \sum_{t=1}^T \mathbb{I}[d_{\omega}(o_{i,t}, a_{i,t}) \geq \tau].$$

SDR uses paired rollouts with shared randomness over a deviation window H :

$$\Delta G_i(t, H) = G_i^{dev}(t, H) - G_i^{contract}(t, H), \quad \text{SDR} = \frac{\#\{(i, t) : \Delta G_i(t, H) > \delta\}}{\#\{(i, t)\}}.$$

B Shock Sweeps and Resilience AUC

B.1 Shock families

We evaluate three shock families: (i) Bernoulli shocks with (p_{shock}, s_{shock}) ; (ii) Poisson shocks with (λ, s_{shock}) ; (iii) Markov shocks with $(p_{01}, p_{10}, s(1))$.

B.2 Sweep grids

Bernoulli:

$$p_{shock} \in \{0.0, 0.05, 0.10, 0.20, 0.30\}, \quad s_{shock} \in \{0.1, 0.3, 0.5, 0.7\}.$$

Poisson:

$$\lambda \in \{0.0, 0.05, 0.10, 0.20, 0.30\}, \quad s_{shock} \in \{0.1, 0.3, 0.5, 0.7\}.$$

Markov:

$$p_{01} \in \{0.02, 0.05, 0.10\}, \quad p_{10} \in \{0.05, 0.10, 0.20\}, \quad s(1) \in \{0.3, 0.5, 0.7\}.$$

B.3 AUC computation

$$\text{AUC} = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} \text{Resilience}(\theta).$$

C Byzantine Stress Tests

Let $\mathcal{B}$ denote Byzantine agents with fraction $f_B = |\mathcal{B}|/n$. A Byzantine agent executes an attack policy with probability π_B and follows the nominal policy otherwise. We evaluate greedy hoarding ($a \equiv 1$) and learned proposer deviations (Appendix D).

D Detector Calibration and ROC/AUC

D.1 Training data

We collect (i) contract samples from π_{contract} labeled $y = 0$ and (ii) deviation samples from proposer q_ψ labeled $y = 1$. Detector is trained with balanced BCE.

D.2 Threshold calibration

We select τ by fixing a target FPR on contract-only data:

$$\tau = \min \{t : \Pr(d_\omega(o, a) \geq t \mid y = 0) \leq \text{FPR}^*\}, \quad \text{FPR}^* = 0.05.$$

We report ROC/AUC [5].

E Silicon Commune Environment

E.1 Dynamics (pseudocode)

Algorithm 2 Silicon Commune Step (pseudocode)

- 1: Inputs: resource R_t , endowments (c_i, m_i) , actions $a_{i,t}$
 - 2: Attempted extraction $E_t \leftarrow \sum_i m_i a_{i,t}$
 - 3: Realized extraction $\tilde{E}_t \leftarrow \min(R_t, E_t)$
 - 4: **for** $i = 1..n$ **do**
 - 5: Allocation $e_{i,t} \leftarrow \tilde{E}_t \cdot \frac{m_i a_{i,t}}{\sum_j m_j a_{j,t} + \epsilon_E}$
 - 6: Utility $u_{i,t} \leftarrow \log(1 + c_i e_{i,t})$
 - 7: Reward $r_{i,t} \leftarrow u_{i,t} - \lambda_c \cdot \text{cost}(a_{i,t})$
 - 8: **end for**
 - 9: Apply shock ξ_t (Appendix B); update $R_{t+1} \leftarrow \max\{0, R_t - \tilde{E}_t + \rho - \xi_t\}$
 - 10: Shape reward via detector (Eq. 3)
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F Hyperparameter Search

We match interaction budgets across methods and run an equal number of trials. We select by validation NSW (or total return when NSW is unavailable).

G Reproducibility Checklist

We will release: (i) environment code, (ii) training and evaluation configs, (iii) all random seeds and sweep scripts, (iv) raw logs for all runs, (v) pretrained checkpoints for main models.